Uses

It is used as a Acidulant in food, (Jams, Preserved fruits, Fruit drinks) and pharmaceutical industries.

1. It is mainly used in food and beverage industry (Jams, preserved fruits, fruit drinks)
2. It is used is pharmaceuticals, and other industrial processes
3. Citrate and citrate esters are used as plasticizers
4. It is used as a chelating and sequestering agent (Tanning of animal skins)

Generally citric acid obtained from citrus fruits, pineapple etc., After the development of microbial fermentation, citric acid production becomes very cheap and easy cost effective.

6.10 Immobilization

It is technique used for the physical or chemical fixation of plant, animal cells, organelles, enzymes or other proteins (monoclonal antibodies) onto a solid matrix or retained by a membrane, in order to increase their stability and make possible their repeated or continued use.

The immobilized enzyme is defined as the enzyme physically confined or localized in a certain defined region of space with retention of its catalytic activity which can be used repeatedly and continuously.

The selection of appropriate carrier and immobilization procedure is very essential procedure is very essential for the immobilization technique.

Various types of materials like cellulose, dextran, agarose, gelatin, albumin polystyrene, Calcium alginate polyacrylamide, collagen carrageenan and polyurethane, inorganic materials (brick, rand, glass, and ceramics, magnetic) are used for immobilization.

The linkage is mediated by ionic bonds, physical absorption or bio specific binding.

The immobilization methods can be classified into four categories

i. Carrier–binding

ii. Cross–linking

iii. Entrapping

iv. Combining

Among all these methods entrapping is discussed in brief.

Entrapping

The enzymes, cells are not directly attached to the support surface, but simply trapped inside the polymer matrix. Entrapping is carried out by mixing the biocatalyst into a monomer solution followed by a polymerization. It is done by change in temperature or by chemical reactions. The diagrammatic representation of entrapping methodology is shown in Figure 6.13.

Advantages of immobilization

1. Immobilized growing cells serve as self proliferating and self regenerating bio catalyst

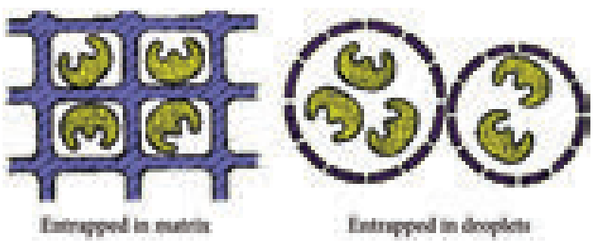


Figure 6.13: Immobilization (Entrapping)

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